

GEORGE, South Africa

WMO: 688280

Lat: 34.0042S

Lon: 22.3842E

Elev: 627

StdP: 14.37

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.3	42.9	25.4	19.9	68.5	30.0	24.7	63.8	27.7	63.3	24.7	64.2	5.2	340	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.9	87.9	64.8	83.8	64.0	80.3	64.3	71.4	78.1	70.1	76.5	69.0	75.0	9.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	111.4	73.7	68.1	105.8	72.6	66.6	100.3	71.2	35.6	78.6	34.5	76.0	33.5	75.4	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.3	20.2	17.5	DB	38.3	97.7	1.3	3.4	37.4	100.2	36.7	102.2	36.0	104.1	35.1	106.6
			WB	35.4	74.7	1.8	1.6	34.2	75.9	33.1	76.9	32.1	77.8	30.9	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	68.0	68.7	66.6	63.4	60.7	57.1	55.8	56.2	57.6	60.5	62.7	66.2	(6)
	DBStd	6.44	3.62	3.89	4.37	5.31	5.16	5.08	5.49	4.76	4.64	4.84	4.09	3.81	(7)
	HDD50	9	0	0	0	0	0	1	6	2	0	0	0	0	(8)
	HDD65	1622	11	8	29	92	157	245	292	278	230	159	92	29	(9)
	CDD50	4362	558	522	515	401	330	214	187	195	230	325	382	503	(10)
	CDD65	501	105	110	79	43	22	8	8	6	9	19	24	68	(11)
	CDH74	2969	391	432	433	368	271	126	111	118	128	207	134	250	(12)
	CDH80	838	69	104	147	137	81	18	16	32	52	90	39	53	(13)
(14) Wind	WSAvg	7.4	7.6	7.2	6.7	6.5	6.5	7.3	7.4	7.6	7.5	8.0	8.0	8.0	(14)
(15) Precipitation	PrecAvg	28.1	2.5	2.2	2.9	2.4	1.8	1.7	1.8	2.6	2.0	3.0	2.8	2.5	(15)
	PrecMax	48.1	8.4	5.6	11.4	11.5	7.8	6.4	6.5	16.8	8.3	9.1	10.1	9.6	(16)
	PrecMin	14.8	0.4	0.6	0.2	0.4	0.1	0.2	0.2	0.4	0.2	0.1	0.5	0.1	(17)
	PrecStd	7.1	1.5	1.1	2.1	2.0	1.7	1.2	1.3	2.6	1.5	2.1	2.1	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	90.0	92.3	91.5	87.7	82.4	82.4	84.4	87.9	90.3	84.5	85.6	(19)
		MCWB	68.4	71.2	68.4	65.3	60.6	56.3	56.5	57.0	61.1	64.1	65.0	69.4	(20)
	2%	DB	80.8	82.3	83.5	84.0	81.3	77.5	77.1	76.7	75.8	78.9	77.1	79.2	(21)
		MCWB	68.7	69.6	67.1	64.2	59.0	55.7	55.1	55.2	59.2	62.8	64.8	67.2	(22)
	5%	DB	77.7	78.6	78.0	77.2	75.5	72.9	72.2	69.8	69.8	72.0	73.2	76.3	(23)
		MCWB	68.3	68.9	67.0	63.3	58.3	54.3	53.9	54.8	58.1	61.3	63.6	66.4	(24)
	10%	DB	75.4	75.6	74.6	72.1	70.6	67.8	67.0	65.0	66.0	68.2	70.1	73.7	(25)
		MCWB	67.2	67.4	66.1	62.2	58.1	53.9	53.1	54.3	56.9	60.5	62.1	65.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	73.4	71.6	69.1	64.7	61.0	59.9	61.9	64.2	67.4	69.2	71.6	(27)
		MCDB	79.2	83.7	81.1	82.1	75.4	70.2	69.4	69.2	76.9	77.8	77.4	79.2	(28)
	2%	WB	71.2	71.4	69.5	66.9	62.6	58.6	57.9	59.4	61.8	64.8	66.8	69.5	(29)
		MCDB	76.4	78.5	77.2	77.3	73.4	68.1	67.7	65.9	70.4	73.3	73.0	75.8	(30)
	5%	WB	69.9	70.0	68.2	65.3	60.9	57.2	56.5	57.6	60.0	63.1	65.1	68.0	(31)
		MCDB	75.3	76.1	75.0	73.4	69.7	65.9	65.7	65.0	67.1	69.9	70.7	73.7	(32)
	10%	WB	68.5	68.8	67.0	63.8	59.7	55.7	55.2	56.1	58.4	61.5	63.5	66.7	(33)
		MCDB	73.6	74.1	73.0	70.2	67.5	64.0	64.1	63.1	64.5	67.2	68.8	72.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	14.9	15.8	17.7	18.9	19.2	18.8	18.6	17.4	16.1	15.3	14.8	(35)
		MCDBR	18.8	19.9	23.4	27.2	27.2	27.7	27.1	28.2	25.6	25.2	21.0	19.0	(36)
	5% WB	MCWBR	9.1	9.4	10.5	12.0	11.2	11.5	11.3	12.6	12.7	12.2	10.6	9.5	(37)
		MCWBR	15.8	16.9	18.5	20.8	21.2	21.5	21.3	21.5	21.8	20.5	18.0	16.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.366	0.369	0.355	0.331	0.316	0.303	0.306	0.333	0.347	0.368	0.355	0.363	(40)
		taud	2.502	2.497	2.523	2.576	2.594	2.601	2.588	2.502	2.446	2.422	2.496	2.497	(41)
	Ebn at Noon		308	300	293	283	271	267	272	278	291	297	309	310	(42)
		Edh at Noon	36	35	32	27	23	22	23	28	34	37	36	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2396	2144	1752	1305	978	836	919	1181	1602	1959	2302	2476	(44)
	RadStd		82	68	84	56	51	47	54	51	75	99	102	134	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page